2 Why Randomize?

This chapter explains why randomized evaluation can help determine what impacts were caused by a program. The modules in this chapter are as follows:

MODULE 2.1: Why It Is Difficult to Measure Causal Impact
MODULE 2.2: Advantages and Limitations of Nonexperimental and Quasiexperimental Evaluation Methods
MODULE 2.3: How Randomization Can Help Infer Cause
MODULE 2.4: Advantages and Limitations of the Experimental Approach

MODULE 2.1 Why It Is Difficult to Measure Causal Impact

Measuring causal impact requires us to compare what happened with the program with what would have happened without the program. This module discusses the fundamental problem of causal inference, namely that we can never observe the same people at the same time both with and without the program: we never directly observe the counterfactual.

What is causal impact?

Causal impact is the difference in outcomes that is *caused by* the program. In other words, to estimate the impact of a program, we need to examine how the people who participated in the program fared compared to how they would have fared if they had not participated in the program. This hypothetical condition is called the *counterfactual.* However, we never observe the counterfactual directly. We observe only

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what happens with the program, not what would have happened in the absence of the program, and so we have to make an inference about the counterfactual. This is the fundamental problem of causal inference. Often we infer the counterfactual from what happened to other people or what happened to the participants of the program before the start of the program. But when we make these inferences, we have to make assumptions: for example, that the people who did and did not participate in the program would have had the same outcomes in the absence of the program. The validity of our estimate depends on the validity of those assumptions.

Example: Did a school uniforms intervention improve learning?

Imagine that we would like to improve educational attainment for children in an area where many children are officially enrolled in school but do not attend regularly and learning levels are quite low. Every two years there is a national survey of the reading and math skills of primary school children. The latest survey finds that half the 8-year-olds in the district we are examining cannot read a simple first-grade paragraph.

We are working with an NGO in the area that wants to help improve educational outcomes and wants to evaluate the impact of giving students school uniforms.[1] In this area, uniforms are not formally required, but there is strong social pressure to wear one, and some children say they are too embarrassed to attend school regularly if they lack a uniform. We think that giving each child a free uniform may help them attend school more regularly and, ultimately, learn more.

The NGO has enough resources to give away 200 uniforms and does so. After two years, the NGO wants to know whether the program worked. Did giving children uniforms help them attend more regularly and learn more? The survey of learning is conducted again, and the reading scores of the children who got the uniforms are higher than they were before the program started. It seems that the program has made a difference. But before we expand the intervention to other schools, we want to know for sure whether the change in reading scores can be attributed to the uniforms intervention, or whether something else caused reading scores to rise. How can we go about finding

1. This example is inspired by a study by Esther Duflo, Pascaline Dupas, and Michael Kremer and is summarized as Evaluation 4 in the appendix.

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out? Is the before-and-after comparison the right comparison for assessing the program?

Many factors (other than a program) can influence changes over time

In the two years between the two learning tests, many factors will have influenced the learning levels of the children who received uniforms. Some are internal to the children in the program: they grow and mature week by week. Even if they never set foot inside a school or open a textbook, children may still learn basic arithmetic from selling trinkets on the street or learn to recognize letters from observing their siblings reading from billboards and labels around them. Program children are also exposed to external events: teachers and parents may change the amount of time and resources they invest in teaching, or the central government may implement a school feeding program in our program schools, which encourages more children to go to school.

All of these events could affect the school attendance and reading scores of the children we are following, and they would do so even if these children were not given uniforms by our program. If we simply measure attendance and reading scores before and after the program, we have no way of telling whether any changes are due to the uniform program or whether some or all of them are due to the many other things that changed and influenced the school attendance and learning of the children in our program during the same time.

Selection matters: Those who sign up for or are chosen for a program are different from those who do not

Instead of comparing reading scores before and after giving children uniforms, we could also compare the reading levels of those children who received free school uniforms to those who did not participate in the program. However, those who receive a program are often different from those who don't. Perhaps we announced our uniforms program and asked parents to sign up. We then gave uniforms to the first 200 children whose parents signed up with the NGO. It's quite possible that the parents who signed up early are more organized or better connected and care more about education than those who did not sign up early.

In other words, these parents differed in ways that would be difficult to observe or measure, and these unobservable differences in parents may also affect how they are able to support their children in

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attending school and learning to read. The more motivated parents may also have the skills and means to send their children to school each day and support their learning by helping them with their homework or by having newspapers around the house. If we compare the scores of those students who got uniforms to the scores of those who did not, we might find that their scores are higher. But this may be due to differences in the motivation of parents, not to our uniforms program.

Alternatively, our program may have targeted the poorest students, who were most in need of uniforms. But the neediest children likely also face other barriers to attending school. In this case, when we compare those who did and did not receive free uniforms, those with uniforms may have lower attendance rates and test scores. Does this mean that our uniforms program kept children from attending school and reduced their scores? There's no good way to tell. It could simply be that children who received uniforms were behind to begin with, and the uniforms were of no help. Or the uniforms may have helped children attend more regularly, but not often enough to catch up with their better-off peers.

We call the tendency for those who receive a program to be different from those who do not get a program selection. Selection can be caused by different people's self-selecting into the program or by the program administrators' allocating the program to particular types of people. If we fail to account for selection in our impact evaluation, we can introduce *selection bias* into our estimate of impact. In other words, we risk attributing differences in outcomes to the program when they are actually caused by preexisting differences in those who self-selected or are selected for the program.

The fundamental problem of causal inference: Finding a valid counterfactual

In the preceding example, after our program was implemented, we would have liked to compare the outcomes of the children who received uniforms with the outcomes of exact clones of these children who did not receive uniforms. In the real world, of course, we can't observe the same group of people both with and without our program at the same time, and there is no way to find exact clones of the people in our program. We can never know what would have happened in the absence of the program. As we said earlier, we never directly observe the counterfactual.

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In Modules 2.2 and 2.3 we discuss how we use different evaluation techniques in an attempt to mimic the counterfactual. All these approaches, however, have to address the fundamental issue that we can never see any one person or any group of people in two conditions at the same time. We therefore end up comparing different groups of people, one group that receives the program (the *treatment group*) and one group that does not receive the program (the *comparison group*). Although there is no exact replica of any specific treatment individual in the comparison group, overall we aim to have similar average group characteristics.

For example, in our uniforms program we are worried that the extent to which parents care about education may affect our outcomes (attendance and test scores) and also be correlated with whether a parent signs up for the program. Every parent is unique, and it will be hard to find two students with parents with exactly the same income levels and degrees of interest in their children's education. Even within families, parental interest may vary between, say, first and later-born children. But if we have many children in the treatment group and in the comparison group, we can find groupings in which the level of income and parental interest in their children's education is the same *on average* between the two groups. All quantitative impact evaluation methodologies have the same aim—to identify treatment and comparison groups that, on average, have similar characteristics—but the methodologies differ in exactly how they identify these different groups.

MODULE 2.2 Advantages and Limitations of Nonexperimental and Quasiexperimental Evaluation Methods

An impact evaluation is only as good as the comparison group it uses to mimic the counterfactual. In other words, the comparison group should mimic what would have happened in the absence of the program. A bad comparison group ruins an evaluation and makes the impact estimate invalid. We discuss a number of different methods of finding a comparison group (the counterfactual) in this module.

Below we discuss the advantages and limitations of some of the most common impact evaluation methods other than randomized evaluations. Randomized evaluations are not the only valid way to conduct impact evaluations. The methods discussed below can be effective—if

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the specific conditions exist that are needed for that method's assumptions to hold. They have the advantage that they can be used retrospectively; in other words, we can go back in time and study the impact of a program that took place years or even decades ago, thus potentially allowing us to measure long-run outcomes even though we have only just started our evaluation. However, the conditions needed for a nonexperimental or quasiexperimental approach to be valid do not always apply. We discuss the assumption required for each evaluation approach below. We focus mainly on quantitative methods of impact evaluation, because these methods are the ones with which we have the greatest expertise, but we start with a short discussion of qualitative methods.

Qualitative impact evaluations

Perhaps the most obvious approach to understanding how a program or policy has changed people's lives is to talk directly to those affected. Qualitative researchers have a wide range of tools for eliciting from participants the impact of a program or policy on their lives. Direct observation can document how individual actions and group dynamics change over time. Open-ended interviews have the advantage of allowing issues to be raised by participants that evaluators might never have thought of and thus might not ask about in closed-ended survey questions. Focus group discussions allow for a pooling of knowledge and experience from a range of community members who may have had different experiences of a policy or program. This interaction between participants can generate insights that no one individual may have. Many qualitative techniques have been developed to help evaluators investigate issues that are painful for individuals to discuss or difficult for them to confront directly, such as experiences of war, corruption, and sexual harassment. For example, participants may be asked to tell a story, draw a picture, or use a physical representation of their experiences or beliefs.

The advantage of the approach

The great benefit of qualitative approaches is the richness of the information collected. For example, rather than just collecting data on the number of days a week a child goes to school, a qualitative analysis can capture the child's level of enthusiasm about school. Qualitative approaches can also document in considerable detail potential mechanisms through which a program may work. Quantitative approaches

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require that this richness be boiled down into a few numbers. This translation into numbers is necessary in order to apply statistical techniques—for example, to say whether the difference between the experiences of two groups is statistically different. But there is no doubt that something is lost in the translation. Even most quantitative evaluators will rely on qualitative techniques at some point. In particular, they are likely to use focus groups and open-ended discussions with participants to generate hypotheses that are then tested using qualitative techniques. When interpreting the findings of a study, quantitative evaluators will often provide quotes from participants or describe in detail the experiences of some individuals. In other words, they will combine quantitative and qualitative evidence. Quantitative evaluators are also increasingly drawing on qualitative techniques to develop quantitative nonsurvey measures of outcomes (we catalog the most popular of these in Module 5.4).

Who is in the comparison group?

Many qualitative studies do not attempt to draw conclusions about the impact of a program or policy. They simply describe a situation. However, qualitative impact studies use a range of approaches to create a comparison group. Often the comparison group is implicit rather than explicit. For example, when participants discuss how the program has changed their lives, the implicit comparison group is the same individuals before the program. Hybrid approaches, however, are emerging whereby qualitative techniques are used to compare treatment and comparison groups (including randomized treatment and comparison groups).

Required data

The difference between qualitative and quantitative approaches is that the latter do not attempt to boil experiences down to data, although most qualitative evaluators will include some data in their assessments.

Required assumptions

If we are to use qualitative approaches to attempt to judge the impact of a program, we must make a number of assumptions. First we must assume that either the evaluator or the participant must have a good understanding of the counterfactual—that is, they must know what would have happened in the absence of the program. When we ask participants how a program changed their lives, we are asking them to

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disentangle all the many changes that were going on in society from those that were driven by the program. This is a hard task, and to some extent we rely on the evaluators to help the participants sort out which of the changes that they associate with the program are in fact general trends and which are program impacts. This, however, assumes that the evaluators know the counterfactual.

The second assumption is either that participants are not influenced by what they think we want to hear or by their own expectations or preferences or that we can see through any biases they may have. There is substantial evidence that people see what they expect to see or want to see. For example, if people expect women to perform poorly, this influences their assessments. Studies have found that the number of women selected for elite orchestras is much lower when auditions are done face to face (and gender is observed) than when those auditioning are behind a curtain (and gender is hidden).[2] Qualitative researchers are trained in techniques that attempt to address these biases. These biases are also an issue for quantitative researchers, and we discuss some solutions in Chapter 5.

Finally, we have to trust that the evaluators have summarized the information in an unbiased way. There are set rules about how to summarize data: we take averages, we look at the level of variation in one group versus another, we measure the level of inequality, and so forth. There is more room for judgment when qualitative evaluators draw conclusions about the impact of a program from the information they have collected. We have to assume that this process is not influenced by the expectations or preferences of the researchers.

Simple before-and-after (pre/post) comparisons

Before-and-after comparisons measure how program participants improved (or changed) over time. The outcomes of the participants are recorded before the program is implemented (baseline outcomes), and these are then compared to the outcomes of these same individuals after the program ends (endline outcomes).

Who is in the comparison group?

Program participants themselves, before participating in the program, serve as the counterfactual.

2. Claudia Goldin and Cecilia Rouse, "Orchestrating Impartiality: The Impact of 'Blind' Auditions on Female Musicians," *American Economic Review* 90 (2000): 715–741.

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Required data

We must collect data on the outcomes of program participants before and after the program is implemented.

Required assumptions

When we use the preprogram outcomes as the counterfactual, we assume that the treatment group's outcomes would have remained the same without the program. We assume that the program was the only factor influencing any changes in the measured outcome over time.

There are many reasons why these assumptions would not hold. As we discuss in the school uniforms example in Module 2.1, many factors other than the program we are evaluating can change the outcomes of program participants over time. These factors include the following:

1. Concurrent changes in the external environment Imagine that at the same time that the free uniform program is implemented there is widespread flooding. The harvest is bad, and children do not have enough to eat. The floods leave a lot of stagnant water, enabling mosquitoes to breed, so children suffer from more malaria, and this affects their ability to go to school and learn. If we compare test scores at the beginning and the end of the year, we will underestimate the impact of the program; we might even conclude that it had a negative impact. If, on the other hand, there is a good harvest or the government introduces a school feeding program that improves school attendance, we would overestimate impact by assuming that all the improvements were due to our program.

2. Concurrent individual changes: Maturation Sometimes it is individuals who change rather than their external environment. These natural changes can lead to changes in outcomes. In our uniforms example, over the year the children will grow older, and as they mature, their cognitive abilities will increase. This maturation can increase their performance on a learning test even if they did not go to school. A before-and-after comparison attributes all changes in learning to the program, thereby overestimating its impact.

3. Rebounding (regression to the mean) Quack doctors around the world have long relied on the fact that most people who are sick get better on their own. If you give them some innocuous potion, you can claim the credit for their recovery. A similar rebounding effect is

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found in communities suffering from war or drought or in individuals suffering from unemployment. Because programs are often conducted in places and with individuals that have been hit by adversity, there is a risk of taking inappropriate credit for natural improvement just as a quack doctor does. A before-and-after comparison attributes any rebound to the program.

Participant–nonparticipant or cross-section comparison

A *cross-section comparison* measures the difference between program participants and nonparticipants after a program is completed.

Who is in the comparison group?

Individuals who didn't participate in the program (for any reason) but on whom data were collected after the program make up the comparison group.

Required data

We must collect data on the outcomes of program participants and nonparticipants after the program is implemented.

Required assumptions

We must assume that nonparticipants are identical to participants except for program participation and that they were equally likely to enter the program before it started (i.e., there is no selection effect).

Selection bias: The problem of preexisting differences

Participant–nonparticipant comparisons alleviate some of the problems of external environment, maturation, and rebounding discussed earlier. However, the people who choose to participate in a program (or who are chosen to benefit from a program) are often systematically different from those who do not. This is called the selection effect.

Let us return to our uniforms example and the case in which widespread floods, poor nutrition, and high malaria rates reduced test scores. Imagine there that are 210 schools in a district in which free school uniforms are distributed, and at the end of the evaluation we compare the outcomes of the schools in the treatment group to those of the schools in the comparison group. If the comparison group was subject to the same floods as the treatment group, the two groups would be comparable in terms of external environment.

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But what if the treatment and comparison groups had different outcomes to start with? Perhaps we sent the free uniforms to the lowest-achieving schools or those in the poorest part of the district, and when we compare program schools to nonparticipating schools it appears that the program *lowered* scores. Or perhaps our program targeted the schools with the most students, but these larger schools also had more resources and a more engaged parent community, which means that their children's test scores were higher to begin with. Now when we compare the scores of participating schools to those of nonparticipating schools, these preexisting differences make it appear that our program *increased* scores.

Multivariate regression

Individuals who received the program are compared with those who did not, and other factors that might explain differences in the outcomes are "controlled" for. In other words, we run a regression of our outcome indicator against treatment status and other indicators that might explain our outcome variable.

Who is in the comparison group?

Individuals who didn't participate in the program (for any reason), but on whom data were collected make up the comparison group.

Required data

In this case we need data not only on indicators of the outcomes of participants and nonparticipants but also on other "explanatory" or "control" variables as well (e.g., age, income levels, education). The control variables need either to be collected at baseline or to be characteristics, like age, that do not change as a result of the program.

Advantage of the approach

Multivariate regression is designed to overcome some of the problems with the simple difference approach outlined above. In our uniforms example, if we are concerned that children in our treatment schools come from slightly poorer households than those in our comparison schools, we can estimate the relationship between income and test scores. We can then factor out any difference in test scores between treatment and comparison groups that is due to differences in income. We then assume that any remaining difference is due to the program.

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We do this by running a regression of test scores on treatment status and on family income. For more details on analysis, including how to run a regression of outcome on treatment status, see Chapter 8.

Required assumptions

We must assume that the control variables we include in our regression include all the relevant ways in which our treatment and comparison groups differ. In particular, we must assume that there are no characteristics that were excluded from the regression (because they are unobservable and/or have been not been measured) that might be different between the two groups and that might affect the outcome variables.

Selection bias: Unobservable differences

The assumption above can be hard to justify. Programs that have some element of self-selection (programs in which people have to decide to participate) are particularly likely to have participants who are different than nonparticipants on traits that are hard to measure, such as motivation or confidence. Yet unobserved traits like motivation are likely to be correlated with many outcomes such as test scores and income. Even when there is no self-selection of who receives a program, we may worry that those deciding where a program should first be introduced had a reason for their decision and that this may lead to unmeasured differences in participants and nonparticipants. At a personal level, we may know the history of how the selection of the program area evolved and feel confident that there are unlikely to be unmeasured differences between participants and nonparticipants. But it can be hard to prove this claim. By their nature, unobservable differences are unobservable, so we cannot prove that there are no unobservable differences between groups with our data. An alternative approach, discussed below under regression discontinuity, is to explicitly exploit the decision rules for introducing a program to create a counterfactual.

Statistical matching

One particular form of multivariate regression is *statistical matching*. Program participants are compared to a group of nonparticipants that is constructed by finding people whose observable characteristics (age, income, education, and so on) are similar to those of the people in the treatment group.

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Who is in the comparison group?

The comparison group can be formed in two ways:

> *Exact matching.* Each participant is matched with at least one nonparticipant who is identical on selected observable characteristics—for example, they have the same gender, age, and occupation.
>
> *Propensity score matching.* Characteristics such as age and income are used to predict who will participate. Every participant and nonparticipant is given a probability or propensity to participate. Each participant is matched with a nonparticipant with the same propensity to participate.

For example, we may find that age, income, education, and past voting habits are correlated with whether or not a person votes. Similarly, educated and wealthier women are more likely to take up microfinance, as are married women. We can calculate the probability that any given woman would take up microfinance if it were made available to her. We can then match a woman who, according to her characteristics and our calculations, had a 60 percent chance of taking up microfinance and did take it up with another woman who also had a 60 percent chance of taking up microfinance but did not take it up.

Why, if these women had the same probability of taking up microfinance, did one take it up and the other not? There are two slightly different approaches to propensity score matching that give slightly different answers to this question. One approach matches those who had access to the program with those who did not have access to the program but had a similar probability of taking it up if they had access. For example, if microfinance is available in some areas but not in others, we match women with a high probability of taking up microfinance in areas with access to microfinance with other women with similar probabilities in an area that does not have access to microfinance.

The other approach to propensity score matching is to find comparators within the group who have access to the program. As discussed below, this approach assumes that the difference in actual take-up is due to chance.

Required data

We need data on outcomes as well as "variables for matching" for both participants and nonparticipants. If we want to match people on

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many different characteristics, we are likely to need large sample sizes and data on lots of variables for matching.

Advantages of the approach

Propensity score matching is a version of multivariate regression. Its advantage is that it puts the most weight on those in the comparison group who are most like those in the treatment group (on observable measures) and who are predicted (based on these characteristics) to be most likely to take up the program. The disadvantage is that it imposes more assumptions than a standard multivariate regression, as we discuss below.

Required assumptions

For this approach to create a valid counterfactual, the set of factors on which we match people must be sufficiently comprehensive that there are no remaining differences between the treatment and comparison groups that might be correlated with our outcome of interest (except for participation in the program). As in the general case of multivariate regression, unobservable factors like motivation are a major concern. We have to assume that we have matched people on enough other characteristics that are correlated with unobservable characteristics that there is no difference left.

Whenever we make this assumption, however, we need to ask ourselves, "If the nonparticipants are just as likely as the participants to take up the program, why didn't they?" Two different approaches to matching answer this question in different ways and require slightly different assumptions. Both of these approaches can be used for exact matching or propensity score matching.

By matching across those with and without access, we find which characteristics predict take-up among those who have access to the program and then find people with those characteristics in an area that does not have access to the program. We compare people with equal probability of take-up in access areas and nonaccess areas. For this comparison to give us an unbiased estimate of the impact of the program, we have to assume that there is nothing that is correlated with access that also has an effect on our outcomes. For example, if microfinance is available in some districts and not in others, we may worry that the districts with access to microfinance are systematically

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different in some unmeasured way from those that do not have access. Maybe the microfinance organization chose to enter districts because they perceived that they had a greater potential for growth. This potential for growth may not be picked up by any of the variables on which we are matching.

When we match between participants and nonparticipants within program areas, everyone in our study is eligible for the program. Thus we worry less about shocks that will affect one group but not the other. But now we are left with the puzzle of why, given that the nonparticipants have the same observable characteristics as the participants, they did not take up the program. The only assumption that will make our evaluation valid is that it was entirely due to chance that some people took up the program while others didn't. Our matching variables are so good that we have explained every difference that could be explained, and chance is the only remaining factor. For example, two women had an equal probability of participating in microfinance, but one happened to pass the microfinance office on her way to visit a friend, whereas the nonparticipant had a friend in the other direction.

Difference-in-difference comparisons

This approach combines a before-and-after comparison with a participant–nonparticipant comparison. It measures changes in outcomes over time of program participants relative to the changes in outcomes of nonparticipants.

Who is in the comparison group?

This group is made up of individuals who didn't participate in the program but on whom data were collected.

Required data

We need data on the outcomes of program participants and nonparticipants both before and after the program is implemented.

Advantages of the approach

This approach deals with the problem of unobservable differences between treatment and comparison groups, as long as those differences have the same effect on outcomes across time. For example, in our school uniforms case we would compare the improvement in test scores for those who did and did not get uniforms. To the extent that

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those parents who signed up early for the uniforms were also more diligent in helping their children with their homework, their test scores were likely to be higher both before the program starts (as measured in a baseline survey) and in our final or endline survey. As long as the effect of having more motivated parents is constant over time, the difference in parental motivation between treatment and comparison groups will not affect the change in test scores. As a result, the comparison of changes in test scores between treatment and comparison groups will be a valid estimate of the impact of the program.

Required assumptions

We must assume that if the program hadn't existed, the two groups would have had identical trajectories over this period. In other words, we have to assume that any differences in characteristics between the treatment and comparison groups do not have more or less of an effect on people's outcomes over time. In our uniforms example, if our treatment group has more motivated and educated parents and if having more motivated and educated parents is more helpful as children get older and their work gets harder, our assumption is invalid. Sometimes we can look at past trajectories of our treatment and comparison groups and check whether these were similar before the program started. This will require several rounds of data collection before the program starts. However, even with this added data it can be hard to convince skeptics that because trajectories in the treatment and comparison groups were similar in the past, their future trajectories would have been similar if not for the program.

Many difference-in-difference designs also require that there be no mismeasurement at the baseline. For example, imagine that we provide a remedial education program to children with low test scores at baseline. If in fact all children have exactly the same level of knowledge but some happen to have received low test scores at baseline because of bad measurement, the scores of these children will automatically bounce back to the average at the endline even if the program does not work. We will incorrectly conclude that the program worked.

Regression discontinuity design (RDD)

This approach is used to evaluate programs that have eligibility cutoffs. For example, children whose parents fall below an income threshold have access to free school meals or those with test scores higher

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than a certain value get into an elite school. Participants who are just above the cutoff are compared to nonparticipants who narrowly missed the cutoff.

Who is in the comparison group?

The comparison group is made up of individuals who are close to the cutoff but fall on the "wrong" side of that cutoff and therefore do not receive the program.

Required data

We need information on exactly how program participation is decided (i.e., where the cutoff is), data on the variables that are used to determine the cutoff (such as income or test scores) for the entire sample, and outcome data. Because we typically use only those participants just above and below the cutoff rather than all participants, RDD works best in evaluating programs with large numbers of participants and nonparticipants.

Advantage of the approach

In other nonexperimental methodologies we have to make assumptions about why people or areas were chosen for the program and hope that there was not some element of this process that introduced unobserved differences between participants and nonparticipants. The advantage of RDD is that if we know exactly what the rules for program allocation are, we can test whether or not they were adhered to, and thus we are less concerned that there is some factor in the allocation that we are not accounting for.

Required assumptions

This approach can answer only questions in cases for which we happen to have a good cutoff that is suitable for the RDD approach. The program rules must introduce a bright discontinuity between those who are eligible and those who are not, and we must be confident that the official criteria for the cutoff were actually followed in practice. (For example, we must be confident the politically powerful or wealthy were not able to game the system and receive the program despite having missed the cutoff.)

We must assume that the cutoff rule was not engineered to produce a certain result (for example, to get a politically favored person or area to just meet the cutoff) and that narrowly meeting or missing

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the cutoff is the only important difference between individuals directly below and directly above the cutoff score. To say this more precisely, we know that those just above the cutoff will be marginally different at the start of the program according to whatever criterion is used for the cutoff, but we must assume that the effect of this marginal difference has only a marginal impact on outcomes.

For example, if we want to evaluate the impact of going to an elite secondary school in Boston or Kenya on future income and we know that admission is based strictly on test scores, we can compare the outcomes of children who just reached the test score cutoff to attend the elite schools and those who just missed the cutoff.[3] Not all those above the cutoff will end up going to the elite schools, but their probability of going jumps up sharply at the test score cutoff. We know that test scores are correlated with future income, but all we need to assume for this approach to be valid is that the relationship between test scores and income would not change sharply exactly at the cutoff in the absence of elite secondary schools. So if we find a sharp jump in the relationship between test scores and future income precisely at the elite school cutoff, we can assume that most of this jump is due to the increased probability of going to an elite school.

One drawback of the approach is that it tests the impact of the program only for those close to the cutoff. In the Boston and Kenyan school examples, we might imagine that the effect of going to an elite school might be different for those who just scraped in and therefore were always toward the bottom of their class compared to those who were average or at the top of their class at the elite school. If we are thinking of expanding a program, the impact on those at the cutoff may be exactly the policy-relevant question.

Instrumental variables

This approach can be used when participation in a program can be predicted by an incidental factor, or "instrumental" variable, that is

3. Adrienne M. Lucas and Isaac Mbiti, "Effects of Attending Selective Secondary Schools on Student Achievement: Discontinuity Evidence from Kenya," working paper, accessed June 1, 2013, http://siteresources.worldbank.org/INTDEVIMPEVAINI/Resources/3998199-1286435433106/7460013-1313679274012/Paper-Lucas_Mbiti_Secondary_201101.pdf; Atila Abdulkadiroglu, Joshua D. Angrist, and Parag A. Pathak, "The Elite Illusion: Achievement Effects at Boston and New York Exam Schools," NBER Working Paper 17264, National Bureau of Educational Research, Cambridge, MA, 2011.

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uncorrelated with the outcome (other than by predicting participation and, through participation, the outcome).

Geographic features can sometimes make good instruments. For example, imagine a remote valley in Indonesia that is separated from a less remote, more densely populated area by a high ridge of mountains.[4] In some parts of the valley, the population can get TV reception designed for the more populated area on the other side of the mountain ridge, and in other parts of the valley the reception is blocked by a particularly high peak on the ridge. Nobody actually crosses the ridge when they want to visit the nearest town; it is much easier to take a different route down the valley and around the ridge. Whether a given community receives TV reception, therefore, is uncorrelated with remoteness or land quality. The height of the nearby ridge affects the population only through its effect on their ability to receive a TV signal. We can then use the height of the ridge to assess the impact of TV on outcomes such as social capital or attitudes toward women.

Who is in the comparison group?

In this case, individuals who are less likely to participate in or benefit from a program or policy because of a nearly random factor are in the comparison group.

Required data

We need data on outcomes, the "instrument," and other control variables.

Advantage of the approach

The advantage of the instrumental variables approach is that it allows us to estimate the causal impact of something that suffers from selection bias by finding an instrument that predicts take-up but is not subject to selection bias. So in our TV example, owning a TV is a choice, and thus comparing people with and without TVs is not a good way to find the causal impact of TV: those who choose to own one are very different from those who choose not to or cannot afford one. But if having a working TV is predicted by something that is not

4. This example is inspired by Benjamin A. Olken's paper "Do Television and Radio Destroy Social Capital? Evidence from Indonesian Villages," *American Economic Journal: Applied Economics* 1 (2009): 1–33.

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a choice, such as the height of the nearest mountain, we can use this to learn about the effect of TV.

Required assumptions

Instrumental variables require a very strong assumption that the instrument affects the outcome only through one specific channel. In our example, the assumption is that the height of the mountain affects social capital or attitudes toward women only through its effect on TV reception. If the height of the nearest mountain affected how long it took to get to the nearest town, it would not be a valid instrument. Nor would it be valid if the height of the nearest mountain affected how much rain the community received, the depth of the river, or the quality of the agricultural land.

It is worth noting that instrumental variables allow us to estimate the effect of a program, policy, or technology only for a very specific group of people. In our TV case, we learn about the impact of TV on those who live in areas where TV reception is dependent on nearby geographic features. We do not learn about the effect of TV in densely populated areas with their own broadcast mast.

Module 2.2 summary

- Qualitative impact evaluations assume that researchers' priorities do not influence the way information is summarized and that beneficiaries and/or the researchers can tell what the counterfactual would have been.
- Before-and-after comparisons assume that any change over time is due to the program.
- Cross-section comparisons assume that any difference between participants and nonparticipants is due to the program and that there is no selection bias (i.e., no systematic differences) between those in the program and those not in the program.
- Multivariate regressions assume that any difference between participants and nonparticipants can be controlled for in a regression. To control for differences, we must be able to measure them.
- Difference-in-difference comparisons correct for differences in starting levels between participants and nonparticipants and compares changes over time between the two groups. We must assume that in the absence of the program there would be no

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difference in the changes over time between participants and nonparticipants.

- Statistical matching sometimes matches those with access to a program with those who do not have access. In this case, matching assumes that there is nothing systematically different between those who do not have access to the program and those who do once we have done our matching. In other words, there was no important reason why some had access to the program and others did not.
- Statistical matching among those with access to the program assumes that (among those who are matched) whether or not someone takes up the program is a matter of random chance.
- Regression discontinuity can be used only when a program has clear rules for eligibility that produce a bright-line cutoff between participants and nonparticipants. It compares those just below and just above the cutoff and assumes that the participation rate is the only thing to jump sharply at the cutoff.
- Instrumental variables require that there be a factor that is not subject to selection bias and predicts participation but affects outcomes only because it predicts participation. The instrumental variable cannot affect anything else of importance through any other channel.

MODULE 2.3 How Randomization Can Help Infer Cause

Randomized evaluations are a type of evaluation that can help us measure causal relationships. This module explains how random assignment helps us create a comparison group that is a valid counterfactual because it is, on average, the same as our group that receives an intervention.

The key feature of a randomized evaluation is that the people who have access to the program or benefit from the policy are selected randomly. This ensures that there are no systematic differences between those who receive the program and those who serve as the comparison group. This is called *random assignment* and differs from *random sampling,* in which people are randomly selected to be surveyed (see Box 2.1). It is this random assignment that gives randomized evaluations an advantage in measuring what impacts were caused by the program.

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BOX 2.1 *Random assignment is not random sampling*

In both random assignment and random sampling we use a random process to select units (individuals, households, schools, etc.). But there is a crucial difference. In random sampling we take a population and use a random process to select units to create a group that is representative of the entire population. We can then measure characteristics of this group and infer from them the characteristics of the entire population. (Most surveys use this approach.) In random assignment we take a study population—a pool of eligible units—and use a random process to assign units to different groups, such as treatment and comparison groups (i.e., we use it to determine access to a program).

All randomized evaluations use random assignment, but most also use random sampling. For example, if we are testing the impact of a program that provides free school uniforms to children in a school, we randomize which school receives the program (random assignment). But we may need to interview only some of the children in each school to get a good picture of impact. (Chapter 6 explains how to determine how many children to interview.) We therefore pick a random sample of children to interview in both treatment and comparison schools (random sampling), and this gives us a good indication of the outcomes of all the children

We may also use random sampling to decide where to perform the evaluation in the first place so that our results will be representative of a wider population (just as a survey is representative of a wider population). This approach is discussed in Chapter 4.

The process of random assignment typically follows a number of steps (illustrated in Figure 2.1):

Define the people who could be eligible for the program. First we define the sample of people from which we will later select the treatment group. It could be that all households in a village are eligible for support. Or our program may be open only to households that fall below a certain poverty threshold. We must have a complete list of all those units (individuals, schools, or clinics) that are going to be in our evaluation before we start the randomization process. In many contexts, this may mean that we need to start by conducting a census of the people in the village or of the schools in the district, because the relevant administrative data may not exist or may be of poor quality.

Randomly assign which units are in the treatment and the comparison groups. From this pool of eligible units, we assign units to

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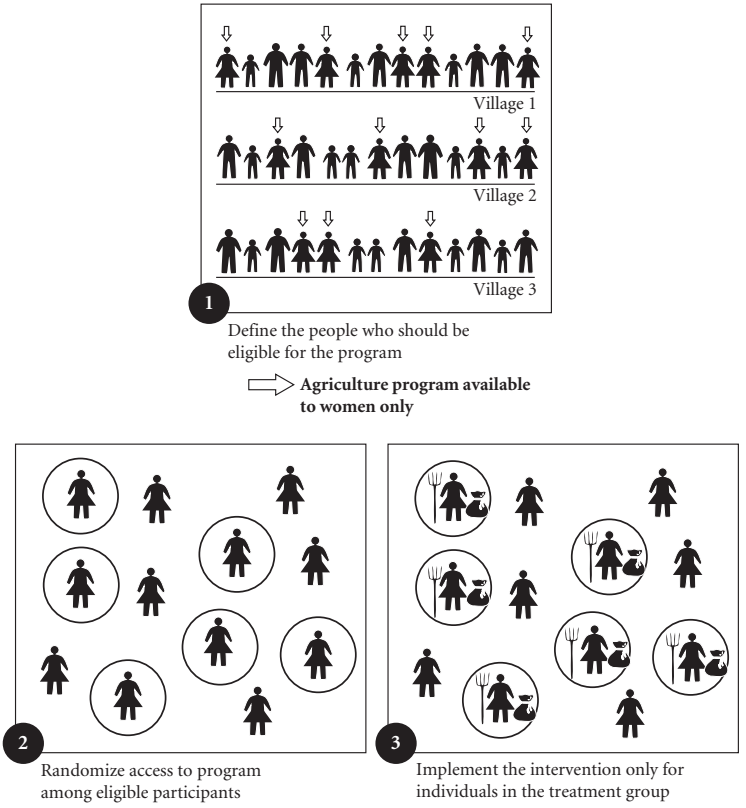

FIGURE 2.1 Steps in the randomization process

different groups by a random process such as a toss of a coin, use of a random number generator, or a lottery. These different groups will then be given access to different types of programs (the treatment groups) or not given access to a program (the comparison group). Chapter 4 provides more details on the pros and cons of different ways of randomizing. The important point is that assignment is purely by chance, so allocation into either the treatment or the comparison groups is completely unrelated to any characteristic that could affect the outcomes. As long as we are working with a large enough number of units, the two groups will be statistically identical. Both will have roughly the same mix of people who are

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motivated to run businesses or are not; more advanced students and those who are less prepared for school; the socially connected and the socially ostracized.

A key advantage of the randomized approach is that in addition to *observable* characteristics (such as income and test scores), *unobservable* characteristics that we don't have a measure of (such as motivation or talent) will also be balanced between the groups. This means that, on average, the treatment and comparison groups will be comparable at the onset of the program, and their trajectory would be the same in the absence of the program.

Implement the intervention only for individuals in the treatment group. After randomization we provide access to the program or the intervention only to individuals, villages, or schools in the treatment group. (For more details on the level of randomization, see Module 4.2.)

Although most people are familiar with the use of randomized evaluations to test the effectiveness of medicines, randomized evaluations also have a long history in agriculture (Box 2.2).

We cannot (and should not) force people to comply with our research protocol. We can only offer the program, and people have every right to refuse for whatever reason. Some will take us up on our offer, and some will not; some in our comparison group will find a way to participate in the program. As we will discuss in greater detail in Chapters 4 and 6, our ability to measure impact will be much stronger if the actual take-up of the program matches our assignment perfectly. However, as long as the proportion of people who actually receive the intervention is higher in the treatment group than in the comparison group, we still have a valid way to measure the impact of the program. (See Chapter 7, which explains ways to help ensure take-up matches assignment as closely as possible.)

MODULE 2.4 Advantages and Limitations of the Experimental Approach

Randomized evaluations answer specific questions in a transparent way without requiring a lot of assumptions. They are designed before a project starts, allowing new, tailored data to be collected. But these charac-

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BOX 2.2 *How did randomized evaluation begin?*

The concept of a control and experimental group was introduced in 1747 by James Lind when he demonstrated the benefits of citrus fruits in preventing scurvy using a scientific experiment.* As a result of his work, Lind is considered to be the father of clinical trials. The method of randomly assigning subjects to comparison and treatment groups, however, was not developed until later.

Randomization was introduced to scientific experimentation in the 1920s when Ronald A. Fisher conducted the first randomized agricultural experiments. Fisher's experimental fieldwork culminated with his landmark book *The Design of Experiments,* which was a key catalyst for the growth of randomized evaluations.

Randomized trials were introduced to government-sponsored social experiments in the 1960s. Rather than being small-scale experiments conducted on plants and animals, these new social experiments were significantly larger in scale and focused on people as the subjects of interest. The model of social experimentation was applied in both Europe and the United States to evaluate programs such as electricity pricing schemes, employment programs, welfare programs, and housing allowances. Since then, social experiments have been used across disciplines and in a variety of settings around the world to guide policy decisions.

*Duncan P. Thomas, "Sailors, Scurvy and Science," *Journal of the Royal Society of Medicine* 90 (1997): 238.

teristics also imply drawbacks: one study can answer only a few questions, and special data collection is expensive. This module describes how understanding these advantages and limitations can help us determine when to use randomized evaluations and when not to.

Tailoring the evaluation to answer a specific question

As we discussed in the previous module, randomizing the allocation of a program or part of a program allows us to determine the causal impact of what we randomize. The advantage of this is that we have precise control over what we randomize and thus what we study. We can randomize the allocation of an entire program, in which case we will learn the impact of the entire program. We can randomize different components of a program, in which case we will isolate the impact of the different parts of the program. We can design our evaluation to precisely test a theoretical concept that economists have worried about for years or to test a contentious policy question. In other words, we can tailor the evaluation to answer exactly the question we need to have answered.

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In the United States, as in many other countries, the poor tend to live in concentrated geographic areas. Many people have speculated that these concentrations of poverty can perpetuate poverty: there tend to be fewer jobs in poor areas, schools may be underfunded and attract less qualified teachers, young people have few successful role models, and crime levels are often high. But it is hard to disentangle the effects of being poor from the effects of being surrounded by other poor people, because the poor who live in disadvantaged areas tend to be different from those who live in richer areas. Researchers worked with the US Department of Housing and Urban Development, which was planning to give housing vouchers to poor families eligible for government housing support, to do this.[5] Some of the families continued to be eligible to receive traditional housing support (access to subsidized housing units in their neighborhoods), while others were given one of two types of vouchers. The first was to subsidize private housing in a neighborhood with less than 10 percent poverty, while the second was to subsidize private housing in a new neighborhood but with no restriction on the poverty level. In other words, the researchers were able to introduce an element of randomization as to whether people lived in poor neighborhoods or not and thus disentangle the effect of poverty on health, employment, and child outcomes from the impact of living in a poor neighborhood on these same outcomes. In this case, the randomized evaluation enabled the researchers to design the evaluation to answer the precise question of interest in a way that would have been difficult to achieve with quasi-experimental approaches.

In Module 4.6 we review a number of examples of other actual randomized evaluations that demonstrate the flexibility of the technique. We show how it can be used to answer underlying questions of human behavior as well as more straightforward questions of whether a program works.

The specificity of the randomized methodology comes with a downside, though. With one treatment group and one comparison group we can answer only one question: what is the impact of the difference in experience we create between these two groups? If we have multiple treatment groups, we can answer more questions, but this greatly

5. This study by Jeffrey Kling, Jeffrey Liebman, and Lawrence Katz is summarized as Evaluation 5 in the appendix.

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increases the cost of the study, and the number of questions we can answer is still very limited.

A contrasting approach is to collect very detailed data on a range of important topics from a representative sample of people across a country or state (examples include the Demographic and Household Surveys and Living Standard Measurement Surveys run by the World Bank, the Micro Sample of the Census or the Current Population Survey in the United States, and their equivalents in other developed economies). In a panel survey, the same individuals or households are interviewed over time: the Indonesia Family Life Survey, the Mexican Life Survey, the Townsend Thai Survey, and the National Longitudinal Study of Youth in the United States are good examples of this approach. These surveys tend to be very large scale and to cover a wide geographic area; often they are nationally representative. They play an important role in describing conditions and trends (which is critical to good program design, as discussed in Module 3.1), and although they are expensive, they have been used to answer many different questions by a wide variety of researchers.

Evaluators often take advantage of quasirandom shocks that have occurred between different waves of large-scale surveys to employ the techniques (discussed in Module 2.2) to evaluate particular policies and programs. For example, the Townsend Thai Survey was used to evaluate the impact of the government's Thai Million Baht Fund, which created locally run microfinance pools in villages across Thailand. Exactly which impact evaluation questions it will be possible to answer with these large, repeated surveys will depend on the shocks that occur, and these may not always be clear at the start. In other words, the advantages and limitations of this approach are the reverse of those of randomized evaluations.

Generalizability: Randomized evaluations answer a question in a specific context

The question of generalizability is often raised with regard to randomized evaluations, and this can take two forms. First, if the experiment is not conducted well, the results obtained may be influenced by the fact that people behave differently when in an experiment or because a different kind of person or organization than is found in the rest of the population agrees to participate in an experiment. We discuss how to design an experiment that minimizes evaluation effects in Module 7.4 and considerations in testing a replicable program in Module 3.3.

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The second and most commonly cited concern about generalizability is whether and under what circumstances the results found in one context apply to other contexts. Unlike the very large national surveys discussed above, randomized evaluations tend to collect data on the specific area in which an evaluation is implemented, which may include a few hundred communities.

The broad question of whether and when results from one evaluation will hold in other contexts is an important one, and we discuss it in detail in Module 9.2. It is worth noting that the question of generalizability is an issue for all evaluations, whether randomized or not. If studying one time and place told us nothing about other times and places, there would be little reason to undertake any evaluation or research. What we need to understand (and can learn by comparing the results of different evaluations) is how and when results from one study help us understand what to do in another situation. As we discuss in Module 9.2, we can generalize only with the help of theory, but our theory can be informed by the evidence we acquire from randomized evaluations.

The need for few assumptions and the transparency of findings

Because we create random variation in exposure to a program or policy, we do not need to make as many assumptions when making causal statements based on a randomized evaluation as we do when we use other approaches to impact evaluation. We do not need to assume that those who did not sign up for a program are just as motivated as those who did sign up. We know that, at least on average, the treatment and comparison groups will be the same in all dimensions at the start of the evaluation.

When we use a simple randomized design, the results of a randomized evaluation can be relatively straightforward to interpret, making the results transparent. As we discuss in Module 8.1, the simplest designs can be analyzed by comparing the mean of our outcome measure in the treatment group with the mean in the comparison group.

The limited number of assumptions and the transparency of the results can provide credibility, which can be useful in convincing policymakers of their relevance. Arguments between researchers about how to interpret a study's results and the sensitivity of those results to different assumptions or different ways to analyze the data can easily turn off policymakers.

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We do not want to overstate this advantage. Not all randomized evaluations are equally simple to analyze or present. We may test many different program variants against each other, which will complicate the analysis. If take-up of the program in the treatment group is low or we cannot find all those we interviewed at the start of the study after the end of the study, we may have to address this problem in the analysis, making it more complex and less transparent. We may also compare the impacts of the program on different types of participants (we discuss the different ways to analyze the results of randomized evaluations in Module 8.2). It is also increasingly common to combine the results from randomized evaluations with statistical modeling techniques. This combined approach allows us to answer more questions, but it does require us to make more assumptions.

For example, a study in Kenya examined the impact of protecting natural springs (encasing each spring in concrete and adding a pipe to bring the spring water to the surface without allowing it to become contaminated).[6] A random sample of springs in the area was chosen for protection. By comparing the health outcomes of those living near the protected springs to the outcomes of those living close to unprotected springs, it was possible to estimate the impact of the program on health. However, the researchers also examined the extent to which people who lived between a protected and an unprotected spring were willing to walk farther to collect their water from a protected spring. By making assumptions about the value of people's time, they estimated a "willingness to pay" for clean water. Using both the actual health benefits of clean water and the willingness to pay for clean water, they constructed an implicit value of a child's life. They also estimated that a private entrepreneur would not find it worthwhile to invest in protecting the springs because they would not be able to charge much for the clean water that was generated.

In other words, by adding a number of assumptions, it is possible to use the results of randomized evaluations to answer a wider range of questions, which has a number of benefits. But when we do this, we give up some of the benefits of simplicity and transparency of analysis.

The results of randomized evaluations can also be less than clear cut if there is debate about the quality of the study. It is certainly the

6. Michael Kremer, Jessico Leino, Edward Miguel, and Alix Peterson Zwane, "Spring Cleaning: Rural Water Impacts, Valuation, and Property Rights Institutions," *Quarterly Journal of Economics* 126 (2011): 145–205.

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case that not all randomized evaluations are of equal quality. However, judging the quality of a randomized evaluation arguably takes less expertise than is needed for some other methodologies. We present criteria to use when assessing the quality of a randomized evaluation in Module 9.1.

Prospective evaluation

Randomized evaluations are nearly always prospective evaluations. In other words, they tend to be designed before the program they are evaluating starts (we discuss a few exceptions below). This generates both advantages and limitations.

Collecting targeted data

Like other forms of prospective evaluation, randomized evaluations allow us to collect detailed data that are tailored to the needs of the evaluation, including data collected before the start of the program (baseline data). Some nonexperimental designs rely on a change in policy as the strategy for identifying causal impact. Because the studies are often conceived after the policy change, the data needed to estimate the impact of the change in policy may not be available. The set of questions that can be answered is limited by the data that can be found (data collected in national surveys that happen to have been conducted at the time of the policy change, for example) or reconstructed ex post. For more discussion on the benefits of having baseline data, see Module 5.2.

Again, however, this benefit comes with a downside: in this case, cost. It is often argued that randomized evaluations are costly, and in many cases they are, with data collection nearly always representing the vast majority of any costs. As a result, the cost of a randomized evaluation will vary greatly depending on the particular data needs of a study. Recent randomized evaluations have ranged in cost from a few thousand to many millions of dollars depending on the length of time over which the study runs, the number of alternative approaches that are tested, whether individuals or whole communities are randomized (see Module 4.2), and the type of outcome measures that are collected.

For example, a study that randomized the price at which pregnant women were willing to purchase insecticide-treated bed nets was relatively cheap. Individual women were randomized to receive nets for different prices, which meant that the sample size for the study

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could be relatively small (see Module 6.2) and geographically concentrated. One of the main outcomes of the study was whether the women purchased the nets, which was observed immediately. Enumerators also went back to individuals' houses to see whether they hung the nets (and whether paying for the nets made purchasers more likely to hang the nets once they had them), but even data on this outcome were relatively easy to collect and were collected shortly after the study started.[7]

There are cases in which randomized evaluations have used data collected by others to assess impact. For example, researchers examining the impact of serving in the Vietnam War were able to use data from Social Security tax records to examine the impact of the draft lottery on earnings.[8] Researchers studying a program of the Colombian government that gave vouchers to attend private schools to randomly selected children were able to use government data on who took the university entrance exam (a good predictor of high school completion) to conduct a relatively inexpensive, long-term follow-up to the study.

Using data that are already being collected by others dramatically reduces the cost of an evaluation, but the downside is that only outcomes on which data happens to have already been collected can be examined. In general, it is rare for data to have been collected on exactly the right individuals and at exactly the right time to be useful as outcome measures for a randomized evaluation, except when the randomization is on a massive scale.

Collaborative design and evaluation

Systematic, creative experimentation, in the tradition of research and development, can lead to innovative solutions. This often requires that the policymaker and the social scientist breach the designer–evaluator distinction and collaborate on the design of the intervention. Because randomized evaluations are prospective and require close cooperation between the evaluator and the implementer, these types of long-

7. This study by Jessica Cohen and Pascaline Dupas is summarized as Evaluation 6 in the appendix.

8. Joshua Angrist, "Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records," *American Economic Review* 40 (1990): 313–336, and "Did Vietnam Veterans Get Sicker in the 1990s? The Complicated Effects of Military Service on Self-Reported Health," Working Paper 09-19, Center for Economic Studies, US Census Bureau, Washington, DC.

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run, in-depth partnerships often develop. Such partnerships can foster the design of new interventions that draw on practical experience, theory, previous impact evaluation results, knowledge of local conditions, and feedback from the local population. By clearly delineating a project as a pilot, the implementing agency may be able to try out new ideas that are more adventurous than they would have been able to try if launching a larger-scale program.

For example, a sequential experiment in Kenya's Western Province examined the relative merits of free distribution and user fees for insecticide-treated (bed)nets (ITNs) in terms of the coverage and use of those nets, which prevent malaria. In the short term, free distribution increases coverage rapidly, but user fees could increase the psychological commitment to using the product. In the long term, free distribution could, in theory, reduce coverage by reducing willingness to buy ITNs. The first experiment examined the impact of price on short-run ITN demand and use, finding that as the price increases, demand falls precipitously and that—contrary to conventional wisdom—as the price increases, usage remains the same.[9] Are there ways to reduce this sensitivity to price, to make people more willing to pay more for ITNs? The second experiment piloted a number of marketing campaigns on the same poor population. None of them had an impact, suggesting that only price matters. But what are the implications of free distribution for long-run coverage? Will people get used to free ITNs and so be less willing to buy them? Or will people learn about the benefits of ITNs and so be more willing to buy them? The third experiment suggests that those who were offered nets for free were more likely to pay for them later, presumably because they had learned that they were useful. This approach of designing new studies that build on the previous ones in the same context is facilitated by being able to tailor the questions very precisely, as we can with prospective evaluations.

Long-term results emerge in the long run

A disadvantage of doing a prospective evaluation is that we have to wait for the program to be implemented, the impact to be felt, and the data to be collected and analyzed before we can assess its impact. How

9. This study by Jessica Cohen and Pascaline Dupas and subsequent follow-up studies by Pascaline Dupas are summarized as Evaluation 6 in the appendix.

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long we have to wait depends on the question we are trying to answer. Some questions can be answered quickly. In other cases, changes in outcomes take a long time to appear (Module 5.2 discusses the timing of data collection).

For example, when researchers wanted to assess alternative approaches to fundraising, they sent fundraisers door to door to collect money for a charity, randomizing the scripts used to solicit the money. In some cases respondents were asked to buy a lottery ticket (with the proceeds going to a charity), in others they were told that their contribution would be matched, and in others they were simply asked for a contribution. The outcome was the amount of money donated, and thus was immediate.[10]

In contrast, an evaluation examining the impact of education and child marriage on health outcomes in Bangladesh randomized a subsidy to families with unmarried adolescent girls (designed to encourage delayed marriage) and educational support to help girls stay in school. The program was effective in delaying marriage and increasing education, but its long-term effects on fertility, complications in pregnancy, and the health of future children will become apparent only over the long term.[11]

Evaluation methodologies that can be used retrospectively (like those that exploit quasirandom shocks that have happened in the past, as discussed in Module 2.2) do not have this limitation. For example, an evaluation of the Rockefeller Foundation's deworming program, which was offered in the US South in the early 1900s, used the timing of program rollout and variations in worm density in different geographic areas to create a counterfactual. The evaluation was done in the early 2000s and was immediately able to look at the long-run impacts of the program.[12] In contrast, a randomized evaluation of deworming in Kenya had to wait 10 years in order to assess 10-year impacts.[13]

10. Craig E. Landry, Andreas Lange, John A. List, Michael K. Price, and Nicholas G. Rupp, "Toward an Understanding of the Economics of Charity: Evidence from a Field Experiment," *Quarterly Journal of Economics* 121 (2006): 747–782.

11. This study by Erica Field and Rachel Glennerster is summarized as Evaluation 7 in the appendix.

12. Hoyt Bleakley, "Disease and Development: Evidence from Hookworm Eradication in the American South," *Quarterly Journal of Economics* 122 (2007): 73–117, doi: 10.1162/qjec.121.1.73.

13. For more details, see the study by Sarah Baird, Joan Hamory Hicks, Michael Kremer, and Edward Miguel, summarized as Evaluation 1 in the appendix.

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Randomized evaluations are not always appropriate or useful

There are many important questions that cannot or should not be answered by a randomized evaluation. In Module 2.2 we have already discussed a range of nonrandomized impact evaluation methodologies and the assumptions under which they are valid. If opportunities arise in which these assumptions hold and a nonrandomized approach will be faster and easier than a randomized evaluation, it makes sense to use these alternative approaches. Other questions can be answered without an impact evaluation at all, or in other cases a randomized evaluation is incapable of answering the question well.

When an impact evaluation is not needed

Many important questions do not require an impact evaluation of any kind. For example, if we find that in a textbook program the textbooks are not delivered or are not used, there is little point in measuring their impact. We cover many other cases in which evaluation approaches other than impact evaluation are sufficient in Module 3.1.

Randomized evaluations are not good at assessing macroeconomic policy

The comparative advantage of the experimental approach is that it can answer specific questions about causal impact. In order to answer them well, we need many units to randomize. But if the policy we want to assess can be introduced only at a national level and if take-up of that policy is immediate for everyone in the country, it will be almost impossible to conduct a randomized evaluation: we would need to randomize some countries to adopt the policy and others not to do so.

An example of such a policy is the adoption of a fixed exchange rate, because there is only one exchange rate for the whole country and everyone is affected by the exchange rate policy. When a policy is introduced nationally but not taken up everywhere, it is still possible to perform a randomized evaluation, as we discuss in Module 4.3. Most macroeconomic questions, however, fall in the category of policies it would be hard—if not impossible—to evaluate using a randomized evaluation.

Randomized evaluations are not good at capturing general equilibrium effects

When an outcome is the result of thousands or potentially millions of interactions by different people, we say that it is established through a

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general equilibrium. An example is the price of gas in Massachusetts. Millions of people in Massachusetts purchase gas in a given week, and thousands of different gas stations sell it. Many factors affect the price, including the world price of gas, local demand, transport costs, the exchange rate, taxes, and refinery capacity. Prices are set through a complex interaction of all these factors. Although prices at one gas station may not be exactly the same as those at another gas station in the same state, there will be a relationship between them. This means that if we shock one part of the system (for example, by randomly raising the price or cutting demand) we will change the whole system (even if only slightly). We will have created what is called a *general equilibrium effect.* Randomized evaluations are not good at capturing these general equilibrium effects because, by definition, the effect is one that is felt at an entire system level, and thus there is no comparison group (at least not within the system). The only way we can use randomized evaluations to look at general equilibrium effects is if we can randomize at the level of the system.

Most wages and prices are established by equilibrating demand and supply between many different purchasers and consumers, and thus any impact of a program on general wages and prices in the area will be hard to capture. This has two implications: first, our estimates from a randomized evaluation may not capture the effect of the program if it were scaled up to the entire population because we have ignored general equilibrium effects, and second, there are important questions that we cannot answer because they are explicitly about general equilibrium effects.

Let us take the example of education and its impact on earnings. We can subsidize the cost of going to school for a random sample of children (maybe by buying them a free school uniform or paying school fees). We can then follow the children we subsidize and those we don't. We may find that those we subsidized get three more years of schooling on average than those we did not subsidize, and they also earn higher incomes later in life. We can conclude that the schooling caused the higher incomes, and we can calculate the impact of schooling on earning. But what would happen if everyone in the country suddenly earned more? Would the wages of educated people fall because there were so many more of them? If so, our evaluation would overestimate the impact of education on earnings if the policy were scaled up. On the other hand, there might be general equilibrium effects that go in the other direction. Perhaps having a concentration

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of more educated people in the country attracts more foreign investment or generates more innovation, which has positive effects on incomes. If this is the case, we will have underestimated the impact of education on earnings if the program were scaled up.

The second issue is that we may specifically be interested in general equilibrium effects themselves. We may want to know whether a large increase in the number of educated workers helps attract more foreign investment. The only way we can measure general equilibrium effects is if markets are segmented—in other words, if smaller local markets are at least partly influenced by local demand and supply. We can then randomize at this local level. For example, a project in France randomized the percentage of local youths that were given job training. They found that helping one young person get a job had a negative effect on the prospect of other young people in the area finding a job.[14] As this example makes clear, general equilibrium effects are a form of spillover—that is, a case in which the impact of the program spills over from those targeted by the program to others. We discuss spillovers and what to do about them in much more detail in Modules 4.2 and 7.3. When markets are somewhat separate from each other, the spillovers are confined to the local area. With a large-scale randomized evaluation we might be able to assess these spillovers, but it will be impossible to measure the effects that are truly general—that is, the ones that affect the entire economy.

The inability to pick up general equilibrium effects is probably one of the most serious drawbacks of randomized evaluations, but it is not unique to them. Anyone using an evaluation methodology that compares one group of people exposed to a program with another group that is not has to worry about general equilibrium issues. It is only when we can compare whole systems that have experienced different shocks that we can measure general equilibrium impacts. The alternative is to build a model of the entire system and make assumptions about how the different parts of the system interact. This is called a general equilibrium model. These models, while relying on a lot of assumptions, are explicitly designed to estimate likely general equilibrium effects.

In the next chapter we discuss in more detail what types of questions can be answered well with impact evaluations and randomized evaluations and which cannot or should not be answered by randomized evaluations.

14. This study by Bruno Crépon, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora is summarized as Evaluation 8 in the appendix.

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Ethical considerations

All evaluation and research that involves human beings raises ethical issues. Randomized evaluations have both ethical advantages and limitations. We start by setting out principles for assessing the ethical implications of any research involving human subjects. It is worth noting that the principles set out here are explicitly for research rather than evaluation. Research is defined (for purposes of these principles) as systematic study that generates lessons that can be applied more widely than in a particular case. In most cases our objective in conducting an evaluation is to draw wider lessons, suggesting that at least for this purpose we can think of our evaluations as research.

Ethical guidelines for evaluation and research

Ethical guidelines exist for researchers working with humans as subjects of their research. Many research organizations have *institutional review boards* (IRBs) that must approve any research proposals involving human subjects before they can proceed (unless the research is deemed to involve minimal risk) and that evaluate proposals against ethical guidelines.

Most ethical guidelines for researchers follow similar principles to those set out in the Belmont Report, which came out of the US National Commission for the Protection of Human Subjects, and we briefly cover the main points here in order to discuss how and when randomized evaluations are compatible with these principles.[15] We strongly encourage all those undertaking research to familiarize themselves with the details of this report and with relevant local guidelines. The three principles in the Belmont Report are the following:

- *Respect for persons.* People's decisions need to be respected. In particular, participants in research must give informed consent to participate in the research. Some groups need to be especially protected because they may not fully understand the risks of participating in research (for example, children) or may find it difficult to refuse to participate (for example, prisoners).

15. The Commission was established in the United States following concerns about unethical research conducted during and after the Second World War. The report can be found at US Department of Health and Human Services, "The Belmont Report," accessed November 28, 2012, http://www.hhs.gov/ohrp/humansubjects/guidance/belmont.html.

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- *Benefice.* Researchers should avoid knowingly doing harm—for example, by encouraging take-up of a program that they have reason to believe is harmful. However, avoiding all risk can be harmful to society in general if it prevents research that could have wide benefits to society as a whole. Thus the overall benefits to society of the research always need to be balanced against the risks. Researchers should seek to maximize possible benefits and minimize possible harm.
- *Justice.* The allocation of risks and benefits between different groups of people should be fair. It is important to avoid a situation in which one group bears all the risk and another stands to reap all the benefits. For example, it would be unjust if a potentially risky vaccine were tested on prisoners and, once approved, were made available only to the rich.

Informed consent

The respect-for-persons principle requires that we tell those whom we would like to take part in a study what the research is about and ask their permission to make them part of the study. This is usually done as part of the baseline survey. The experiment is explained, and participants are asked for their permission to continue. Depending on the level of risk, either oral or written consent is required. Oral consent can often be beneficial, because it allows the illiterate (who might otherwise be too intimidated to participate) to be included in the program and the study, but usually special permission is required from the illiterate. In some situations, if risks are minimal and getting informed consent would either be very difficult or undermine the validity of the experiment, the informed consent rules can be waived or participants can be given partial information. An example of this would be a study in which explaining that an experiment was looking at participants' levels of racial bias might change how people answered questions.

There can be some ambiguity over informed consent rules when only a subset of those who take up an evaluated program is surveyed for the study. This arises mainly when a whole community is randomized to gain access to the program. In some cases those not surveyed are not considered part of the study, and thus informed consent from them is not required. Usually, although access is open to the whole community, individuals still have a choice as to whether to participate, and the decision to take up the program can be considered informed

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consent (as long as the implementer carefully explains the program to those who decide to take it up). If, however, take-up is not voluntary (for example, if fluoride or chlorine are added to the municipal water source), it becomes particularly important for the entire community to be consulted and any risks explained. Even when take-up is voluntary, it is good practice to inform the entire community about the study and the program in cases in which the entire community is randomized to receive access to a program.

Evaluating questions of relevance to the population being studied

Usually social science research is designed to answer questions relevant to the population being studied. For example, we study job counseling for unemployed youths in France because we want to improve job prospects for youth in France. We study the take-up of bed nets for pregnant women in the Western Province of Kenya because we want to design policies for increasing the use of bed nets by pregnant women in that area. This enables us to comply with the justice principle.

Do evaluations deny access to a program?

But what about the benefice principle? Don't randomized evaluations do harm because they lead to people being denied access to a program from which they would otherwise have benefited? And don't they involve testing programs whose benefits and risks are unknown on the poor? Before addressing these questions it is worth pointing out that these two concerns are in direct opposition to each other. The concern that randomized evaluations deny access to a program makes the assumption that we know a program is beneficial. In most cases, however, we do not know the impact of the program—that is why we are evaluating it. The concern that we are testing an unknown program implies we should "deny" access to everyone in case the program has negative effects. The case that research, and in particular randomized evaluations, are ethical comes from the position that if we do not know whether a program is beneficial, society benefits if its impacts are tested before it is scaled up to more people. There may be some risk to those who experience the program first, but those risks need to be balanced against the potential benefits of better understanding its impacts; those risks are explained to participants, and they accept them voluntarily.

Usually conducting a randomized evaluation neither increases nor decreases the number of people who are likely to receive a program

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even in the short run; it simply changes the geographic spread of the program. Without the evaluation, the program might be delivered to all eligible people in one district, but with it the program may be delivered to half the people in two districts. Sometimes changing who receives the program may have ethical implications: for example, if a program is very carefully targeted to the neediest, evaluating it with a randomized evaluation may require finding a larger, less targeted pool. We discuss the ethical issues in this case and how to design an experiment that minimizes harm in more detail in Module 4.3, in which we consider a particular randomization design known as a treatment lottery around a cutoff.

Weighing the risks and benefits

It can even be ethical to perform a randomized evaluation if doing so reduces the number of people receiving a program in the short run (for example, because wide-scale rollout is delayed to allow a pilot to be evaluated). Again, we need to weigh the likely risks and benefits. We have to weigh the potential harm caused to those who would have received the program earlier had we not done the research against the potential benefit from having rigorous evidence on its impact. The benefits include being able to use evidence of a positive impact to raise more funding to expand the program in the long run if it is effective or being able to replace the program with something more effective if its impact is disappointing. The evaluation may avoid harm if we find that the program has unintended negative consequences and it is not scaled up as a result.

For example, a program run by ICS designed to strengthen local women's self-help groups by providing training and finance in Kenya's Western Province was evaluated through a randomized evaluation. The results showed that the treatment groups were more likely to be taken over by richer, more educated women, while the most vulnerable women ceased to be members.[16] The evaluation helped ICS avoid scaling up a program that everyone had thought would be beneficial to vulnerable women but that had arguably hurt the interests of some of those it was designed to help.

16. Mary Kay Gugerty and Michael Kremer, "Outside Funding and the Dynamics of Participation in Community Associations," *American Journal of Political Science* 52 (2008): 585–602.

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In weighing costs and benefits, an important consideration is how much evidence we have on the likely benefits of the program. If we already have good reason to think that a program will be beneficial, this means the bar for the benefits of doing the research must necessarily be higher.

Finally, it is worth noting that it can be ethical for researchers to evaluate a program they may be concerned has harmful effects, although the criteria under which such studies are approved are tightly regulated and the potential benefits of these studies need to be high. Imagine that there is a program that is quite commonly implemented but that the researcher is concerned may have harmful effects. The researcher does not have enough evidence to convince those running the program that it is harmful and should be shut down. It might well be that the benefits of gathering evidence of harm through a research program would outweigh the potential risks to those who take part in the research. The key criteria for moving forward ethically in such a context would be that there is no hard evidence that the program is harmful (if there were, we could use that evidence to have the program shut down without the evaluation), that all those taking part in the study will be warned of the potential risks, and that the evidence generated is likely to be effective in reducing the prevalence of the program in the future if the results show it causes harm.

Cross-cultural views on what is ethical

The ethical guidelines briefly summarized above were formulated in the United States. If we are running a randomized evaluation in another country, it is important to take into account views of what is ethical in that country. Many countries outside the United States have their own ethical guidelines and review processes, either at the national level or within individual institutions, for research involving human subjects. These can help researchers take into account local cultural ethical standards. For the most part, the guidelines these institutions follow are similar to US guidelines, although in some cases they are more explicitly medical in nature and less appropriate to the needs of social scientists. In some countries IRBs for research involving human subjects cover only medical studies, and in others the coverage is unclear. Even if there is no local review board or no review board that covers social science evaluations, it is important to understand which approaches would be considered ethical locally and which would not.

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In the places J-PAL researchers have worked, using randomization or a lottery to share scarce resources is in general seen as an appropriate and ethical approach. Indeed, in many cases beneficiaries see randomization or a lottery as fairer than their typical experience, which may include allocation based on political connections, the magnitude of a bribe, or some unexplained and seemingly arbitrary decision-making. However, it is important to make sure that randomization is done in an appropriate way. For example, it may be that randomly choosing some children in a class to receive a free lunch while not giving a free lunch to others in the same class may be considered inappropriate, while choosing some schools to receive a free lunch while not giving it to all schools is considered quite normal.

We discuss how ethical issues should be incorporated into the decision on whether to randomize at the group or the individual level in Module 4.2 and the particular ethical issues related to different forms for randomized evaluations in Module 4.3.[17]

17. For further reading on the ethics of randomized evaluations, see Rachel Glennerster and Shawn Powers, "Assessing Risk and Benefit: Ethical Considerations for Running Randomized Evaluations, Especially in Developing Countries," in *Handbook on Professional Economic Ethics* (Oxford, UK: Oxford University Press, forthcoming).

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